

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Case Study (Medium)

Assessment Tool Weightage: 30%

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QUESTIONS Total Marks: 50			
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5

6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5

8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

*I **KESHA VAARAM L/192511253**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

L. Kesha

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly

Applicati on of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understand abl e presentatio n	Limited clarity and structure	Poor clarity; difficult to understand

QUESTION 1 — ONLINE SHOPPING PLATFORM

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)
Task: Identify functional dependencies and normalize the relation to 3NF.

2.1 Business rules and assumptions

- Each OrderID identifies one order.
- Each CustID identifies one customer and therefore determines CustName.
- Each ProdID identifies one product and therefore determines ProdName and Price.
- An order may contain multiple products, so an order-product combination identifies Qty.

2.2 Functional dependencies

- OrderID → CustID, CustName
- CustID → CustName
- ProdID → ProdName, Price
- (OrderID, ProdID) → Qty

(OrderID, ProdID) is the natural candidate key for the line-item relation.

2.3 Normalization

The original relation mixes order facts, customer facts, product facts and order-line facts. CustName is transitively dependent on OrderID through CustID, while product details depend only on ProdID.

CUSTOMER	CustID, CustName	CustID
PRODUCT	ProdID, ProdName, Price	ProdID
ORDER_HEADER	OrderID, CustID	OrderID
ORDER_LINE	OrderID, ProdID, Qty	(OrderID, ProdID)

Foreign-key relationships: ORDER_HEADER.CustID references CUSTOMER.CustID; ORDER_LINE.OrderID references ORDER_HEADER.OrderID; ORDER_LINE.ProdID references PRODUCT.ProdID.

The online-shopping relation is decomposed into CUSTOMER, PRODUCT, ORDER_HEADER and ORDER_LINE. Customer and product facts are stored once, eliminating major update, insertion and deletion anomalies and producing a 3NF design.

3. QUESTION 2 — HOSPITAL PATIENT SYSTEM

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)
Task: Identify anomalies and normalize the relation to BCNF.

3.1 Assumed dependencies

- PID → PName, DocID, WardNo
- DocID → DocName, Specialization
- WardNo → WardName

3.2 Anomalies

Update	Changing a doctor's specialization or ward name requires repeated updates across many patient records.

Insertion	A new doctor or ward cannot be represented cleanly if no patient is currently assigned.
Deletion	Deleting the last patient assigned to a doctor or ward may accidentally remove the only stored doctor/ward information.

3.3 BCNF decomposition

PID is a key of the patient relation. However, DocID and WardNo are determinants that are not superkeys of the original relation, so the original relation violates BCNF.

PATIENT	PID, PName, DocID, WardNo	PID
DOCTOR	DocID, DocName, Specialization	DocID
WARD	WardNo, WardName	WardNo

In PATIENT, PID determines the remaining attributes. In DOCTOR, DocID is the determinant and key. In WARD, WardNo is the determinant and key. Thus every non-trivial FD in each resulting relation has a superkey on the left side.

The hospital schema is decomposed into PATIENT, DOCTOR and WARD. The decomposition removes the non-key determinants from the original relation and satisfies BCNF under the stated assumptions.

4. QUESTION 3 — UNIVERSITY ENROLLMENT

Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade) Task: Analyze functional dependencies and apply 2NF decomposition.

4.1 Functional dependencies

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $(SID, CourseID) \rightarrow Grade$

(SID, CourseID) is the candidate key.

4.2 Identify the 2NF violation

Because the key is composite, a non-prime attribute must not depend on only part of the key. Here SName depends only on SID, while CourseName and Faculty depend only on CourseID. These are partial dependencies.

$SID \rightarrow SName$	SName depends on only one part of (SID, CourseID).
$CourseID \rightarrow CourseName, Faculty$	Course details depend on only CourseID, not the complete enrollment key.

4.3 2NF decomposition

STUDENT	SID, SName	SID
COURSE	CourseID, CourseName, Faculty	CourseID
ENROLLMENT	SID, CourseID, Grade	(SID, CourseID)

The decomposition removes all partial dependencies. STUDENT stores student facts, COURSE stores course facts, and ENROLLMENT stores the student-course relationship and grade. The resulting design is in 2NF.

5. QUESTION 4 — AIRLINE BOOKING

Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)
Task: Normalize the airline booking schema up to BCNF.

5.1 Business rules

- PassID identifies a passenger and determines PassName.
- A FlightNo identifies a scheduled flight route: DeptCity and ArrCity.
- A particular flight on a particular Date can contain multiple passengers.
- A passenger has one seat for a particular flight-date booking.

5.2 Functional dependencies

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity}, \text{ArrCity}$
- $(\text{FlightNo}, \text{PassID}, \text{Date}) \rightarrow \text{Seat}$

The composite key for a booking record is taken as $(\text{FlightNo}, \text{PassID}, \text{Date})$.

5.3 BCNF-oriented decomposition

PASSENGER	PassID, PassName	PassID
FLIGHT	FlightNo, DeptCity, ArrCity	FlightNo
BOOKING	FlightNo, PassID, Date, Seat	$(\text{FlightNo}, \text{PassID}, \text{Date})$

Each determinant in the resulting relations is a key of its own relation. Therefore, under the stated business rules, the decomposition satisfies BCNF.

If the real airline system allows FlightNo to be reused across different dates with different routes, then the flight key/business rules must be changed. The normalization result depends on the actual business rules.

The airline booking relation is normalized to PASSENGER, FLIGHT and BOOKING, removing passenger and flight transitive/partial dependencies and producing a BCNF-oriented design.

6. QUESTION 5 — MULTINATIONAL HR SYSTEM

The brief asks for an HR schema analysis but does not specify the exact attributes. The following explicit case schema is used for a complete worked solution: EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName, CountryID, CountryName, Salary)

6.1 Functional and transitive dependencies

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}, \text{ManagerID}, \text{CountryID}, \text{Salary}$

- DeptID → DeptName, ManagerID
- ManagerID → ManagerName
- CountryID → CountryName

EmpID → DeptID → DeptName

EmpID → ManagerID → ManagerName

EmpID → CountryID → CountryName

6.2 Why normalization is needed

Department, manager and country information is repeated for every employee belonging to the same department, manager or country. This creates update, insertion and deletion anomalies.

6.3 3NF design

EMPLOYEE	EmpID, EmpName, DeptID, ManagerID, CountryID, Salary	EmpID
DEPARTMENT	DeptID, DeptName, ManagerID	DeptID
MANAGER	ManagerID, ManagerName	ManagerID
COUNTRY	CountryID, CountryName	CountryID

The assumed multinational HR schema is decomposed to 3NF by separating employee, department, manager and country facts. The transitive dependencies are removed from EMPLOYEE.

7. QUESTION 6 — LIBRARY SYSTEM & 4NF

Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)
Task: Identify MVDs and decompose to 4NF.

7.1 Important modeling observation

The supplied schema represents borrowing transactions, so BorrowDate is naturally associated with a particular member-book occurrence. Multi-valued dependencies arise if a member can independently have multiple books and multiple independent hobbies/preferences or other independent multivalued facts. The given attributes alone do not provide a second

clearly independent multivalued attribute.

To demonstrate the requested 4NF case study, assume Genre represents an independent multi-valued classification associated with a book and that a book can have multiple authors independently of its genres. This yields the MVDs BookID $\twoheadrightarrow$ Author and BookID $\twoheadrightarrow$ Genre.

7.2 MVDs

- BookID $\rightarrow$ Title
- BookID $\twoheadrightarrow$ Author
- BookID $\twoheadrightarrow$ Genre

7.3 Redundancy illustration

B01	Author A	Science
B01	Author A	Technology
B01	Author B	Science
B01	Author B	Technology

Independent author and genre sets generate combinations. Storing all combinations in one relation creates multivalued redundancy.

7.4 4NF decomposition

BOOK	BookID, Title	BookID
BOOK_AUTHOR	BookID, Author	(BookID, Author)
BOOK_GENRE	BookID, Genre	(BookID, Genre)
BORROW	MemberID, BookID, BorrowDate	(MemberID, BookID, BorrowDate)

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Under the explicit assumption, the independent multi-valued facts are separated into BOOK_AUTHOR and BOOK_GENRE, while borrowing is kept as a transaction relation. This removes the 4NF redundancy.

8. QUESTION 7 — TELECOM BILLING

The brief asks for all candidate keys, primary keys and FDs, but does not provide the exact telecom schema. The following worked case is used:

BILLING(InvoiceNo, CustomerID, CustomerName, PlanID, PlanName, BillingMonth, Amount)

8.1 Assumed functional dependencies

- InvoiceNo $\rightarrow$ CustomerID, PlanID, BillingMonth, Amount
- CustomerID $\rightarrow$ CustomerName
- PlanID $\rightarrow$ PlanName
- (CustomerID, BillingMonth) $\rightarrow$ PlanID, Amount

8.2 Candidate keys

InvoiceNo⁺ = {InvoiceNo, CustomerID, CustomerName, PlanID, PlanName, BillingMonth,

Amount} Therefore InvoiceNo is a candidate key and can be selected as the primary key.

Under the stated assumptions, (CustomerID, BillingMonth) also determines PlanID and Amount, and CustomerID determines CustomerName while PlanID determines PlanName. Therefore it is an alternate candidate key if the business rule guarantees one invoice per customer per billing month.

Candidate key	InvoiceNo	Uniquely identifies the invoice.
Candidate key / alternate	(CustomerID, BillingMonth)	Uniquely identifies one bill per customer month under the stated rule.
Primary key	InvoiceNo	Chosen as the principal identifier.

8.3 Key observation

Candidate keys must be minimal superkeys. A primary key is one candidate key selected for implementation; other candidate keys can be enforced with UNIQUE constraints.

The telecom case demonstrates how business rules determine candidate keys. InvoiceNo is the primary key, while (CustomerID, BillingMonth) can serve as an alternate candidate key if the stated uniqueness rule is valid.

9. QUESTION 8 — RETAIL INVENTORY SYSTEM

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)
Task: Apply complete normalization.

9.1 Functional dependencies

- PID → PName, SupplierID, Category, Price, Warehouse
- SupplierID → SupplierName

PID → SupplierID → SupplierName

9.2 Normalization analysis

The relation is assumed to be in 1NF because all listed attributes are atomic. Since PID is a single-attribute key, there is no partial-dependency problem for 2NF. However, SupplierName is transitively dependent on PID through SupplierID, so the original relation is not in 3NF.

9.3 Complete normalized design

1NF	Ensure each product row contains atomic values.
2NF	Satisfied because the candidate key PID is single-attribute.
3NF	Remove SupplierName from PRODUCT and create SUPPLIER.
BCNF	Each determinant in the resulting relations is a key of its relation under the stated FDs.
4NF / 5NF	No additional non-trivial MVD/JD is specified, so no further decomposition is justified from the given case.

PRODUCT	PID, PName, SupplierID, Category, Price, Warehouse	PID

SUPPLIER	SupplierID, SupplierName	SupplierID
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The complete normalization process stops at the BCNF-level design supported by the stated dependencies: PRODUCT and SUPPLIER. Further 4NF or 5NF decomposition should not be invented without additional MVDs or join dependencies.

10. QUESTION 9 — SCHOOL ERP

The brief does not provide the exact poorly normalized School ERP schema. The following representative schema is used: SCHOOL(SID, SName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName, Grade)

10.1 Functional dependencies

- $SID \rightarrow SName, ClassID$
- $ClassID \rightarrow ClassName, TeacherID$
- $TeacherID \rightarrow TeacherName$
- $SubjectID \rightarrow SubjectName$
- $(SID, SubjectID) \rightarrow Grade$

10.2 Normalization problems

The candidate key is (SID, SubjectID). SName depends only on SID, while SubjectName depends only on SubjectID; these are partial dependencies. ClassName and TeacherName are transitively dependent on SID through ClassID and TeacherID.

10.3 BCNF-oriented normalized design

STUDENT	SID, SName, ClassID	SID
CLASS	ClassID, ClassName, TeacherID	ClassID
TEACHER	TeacherID, TeacherName	TeacherID
SUBJECT	SubjectID, SubjectName	SubjectID
ENROLLMENT	SID, SubjectID, Grade	(SID, SubjectID)

10.4 Justification

- STUDENT stores only student facts and class reference.
- CLASS stores class facts and its teacher reference.
- TEACHER stores teacher facts.
- SUBJECT stores subject facts.
- ENROLLMENT stores the many-to-many student-subject relationship and grade.

The representative School ERP is decomposed into independent entity and relationship tables. Under the stated dependencies, each determinant is a key in its relation, giving a BCNF-oriented design with reduced redundancy.

11. QUESTION 10 — MOVIE STREAMING PLATFORM & 5NF

The brief asks for join dependencies and a 5NF decomposition with lossless-join verification but does not specify the exact schema. The following case is used:

STREAM(MovieID, ActorID, PlatformID)

11.1 Business rule and join dependency

Assume a movie is independently associated with a set of actors and a set of streaming platforms, and the valid combinations are implied by three binary relationships: Movie–Actor, Movie–Platform and Actor–Platform.

***(MA, MP, AP)**

where MA(MovieID, ActorID), MP(MovieID, PlatformID), and AP(ActorID, PlatformID).

11.2 Redundancy example

M01	A01	P01
M01	A01	P02
M01	A02	P01
M01	A02	P02

If the same facts are represented by the three binary relationships, storing every triple duplicates information. A non-trivial join dependency indicates that the original ternary relation can be reconstructed from smaller relations.

11.3 5NF decomposition

MOVIE_ACTOR (MA)	MovieID, ActorID	(MovieID, ActorID)
MOVIE_PLATFORM (MP)	MovieID, PlatformID	(MovieID, PlatformID)
ACTOR_PLATFORM (AP)	ActorID, PlatformID	(ActorID, PlatformID)

11.4 Lossless-join verification

Under the assumed join dependency, the original STREAM relation is exactly the join of the three decomposed relations: **STREAM = MA ⋈ MP ⋈ AP**

Therefore, joining the decomposed relations recreates the permitted movie-actor-platform combinations without requiring an additional independent fact. This is the lossless reconstruction condition for the assumed join dependency.

A relation is in 5NF when every non-trivial join dependency is implied by candidate keys. The decomposition removes the explicitly assumed non-trivial join dependency and stores the component relationships separately.

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The movie-streaming case is decomposed into MOVIE_ACTOR, MOVIE_PLATFORM and ACTOR_PLATFORM. Under the stated join dependency, the original relation can be losslessly reconstructed by joining the three relations.
